



NCIE-010

Evidence, Provenance & Audit Architecture

**In development — for human review of full expansion —
Version 1.0**

Specifies Evidence lifecycle and chain-of-custody engineering plus enterprise provenance and audit architecture beneath the approved NCIE-001–009 foundation: Evidence object model, acquisition, integrity, custody, versioning, invalidation, relationships and contradiction preservation, annotation, derived-evidence lineage, AI/Agent/ARGUS provenance boundary, temporal semantics, Evidence-to-Finding traceability, provenance graph, audit-event architecture, security/classification, retention interface, search provenance, reconciliation/recovery and verification/assurance testing.

Evidence-only, governance- and security-consuming: NCIE-010 owns Evidence integrity, lineage, provenance, accountability, audit and defensibility, but does not redefine NCIE-003 canonical entities, NCIE-008 governance authority, NCIE-009 security/IAM controls, NCIE-006 Memory semantics or NCIE-014 physical storage.

Expands approved NCIE-010 Skeletal v0.1, itself traced to approved NCIE-001 v1.3, approved NCIE-002 v0.5, approved NCIE-003 v0.4, approved NCIE-004 v1.2, approved NCIE-005 v1.3, approved NCIE-006 v1.2, approved NCIE-007 v1.2, approved NCIE-008 v1.0 and approved NCIE-009 v1.0.

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Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

0.1 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture. NCIE-010 owns Evidence lifecycle/chain-of-custody engineering plus enterprise provenance/audit architecture; it creates no canonical-entity meaning, no institutional decision right and no security control of its own.

0.2 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–009 authority the chapter builds on, stating only the Evidence/provenance/audit relationship rather than restating upstream technical, governance or security content.

0.3 Governing Evidence/Provenance/Audit Invariants Register

These 19 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-010 Skeletal v0.1 and by the approved NCIE-001–009 authority chain; no institutional decision is required to adopt them. The 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants and other upstream invariants already binding from NCIE-005/006/007 remain fully in force and are cross-referenced, not restated, in the chapters below.

- SOURCE DATA ≠ EVIDENCE.
- MEMORY ≠ EVIDENCE.
- AI / AGENT OUTPUT ≠ EVIDENCE.
- EVIDENCE REFERENCE ≠ EVIDENCE CONTENT.
- EVIDENCE REGISTRATION ≠ FINDING.
- EVIDENCE SUPPORT ≠ INSTITUTIONAL DECISION.
- PROVENANCE ≠ AUTHORITY TO DISCLOSE.
- LINEAGE ≠ CAUSALITY.
- AUDIT EVENT ≠ PROVENANCE RECORD.
- TELEMETRY ≠ AUDIT ≠ PROVENANCE.
- ANNOTATION ≠ ALTERATION OF SOURCE EVIDENCE.
- CORRECTION ≠ HISTORICAL ERASURE.
- INVALIDATION ≠ DELETION OF HISTORICAL RECORD.
- CONTRADICTION ≠ ERROR TO BE SILENTLY REMOVED.
- CHAIN OF CUSTODY ≠ CONTENT OWNERSHIP.
- HASH / INTEGRITY METADATA ≠ TRUTH OF CONTENT.
- VPF PASS ≠ EVIDENCE.
- FINDING ≠ DECISION.
- DECISION-TIME RECONSTRUCTION ≠ CURRENT-STATE REWRITE.

0.4 Evidence/Security/Governance Boundary Note

NCIE-010 owns Evidence lifecycle and chain-of-custody engineering plus enterprise provenance/audit architecture. It consumes NCIE-003 canonical Evidence/Source Object meaning, NCIE-008 institutional decision rights, and NCIE-009 security/IAM/classification controls, without redefining any of them. Where a matter requires an institutional choice (an admissibility standard, a retention period, a custody-attestation threshold) not fixed by an approved source, this document states the control class and defers the choice to Human Review rather than inventing an answer. Approval of the NCIE-010 Skeletal v0.1 authorized this full expansion only; it did not itself approve any control default, threshold, product/algorithm choice or ownership assignment proposed below.

0.5 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 57, a Chapter 28 register item that restates an identical NCIE-008/009 governance or security question cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-010 item is created only for a distinct Evidence/provenance/audit decision. Chapter 28 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §28.4).

0.6 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual control default, threshold, product choice or algorithm proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Nothing in this document may be cited as an approved institutional decision, a deployed control, an evidentiary admissibility standard, or a certification.

0.7 Salient Things to Note Discipline

Directive item 70 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 29 chapters and explicitly names the NCIE-009 Chapter-27 omission (found by independent human examination and corrected before this document was produced) as the failure mode not to repeat. Every chapter below therefore renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-010's lib.py raises a build error if a chapter's salient content is missing, rather than being added and checked afterward), and Chapter 29's mechanical audit (§29.2) re-verifies this against the final document.

0.8 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-010 Skeletal v0.1 into the complete 29-chapter specification per the NCIE-010 v0.1→v1.0 Claude Full-Expansion Production Directive: Evidence/provenance/audit doctrine, five consolidated matrices, threat/contradiction-preservation audits, Human Review Register (Chapter 28) and full traceability/handover (Chapter 29).

Authority: Approved NCIE-001 Master PRD v1.3, approved NCIE-002 System Architecture & Technical Design v0.5, approved NCIE-003 Data Model & Data Dictionary v0.4, approved NCIE-004 Technology Stack & Engineering Blueprint v1.2, approved NCIE-005 ARGUS Intelligence Assistant Specification v1.3, approved NCIE-006 Memory & Context Architecture Specification v1.2, approved NCIE-007 AI, Agent & Model Orchestration Specification v1.2, approved NCIE-008 Governance, Authority, Accountability & RACI Specification v1.0, approved NCIE-009 Security, IAM & Zero-Trust Architecture v1.0, and the approved NCIE-010 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 29-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

Document	NCIE-010 Evidence, Provenance & Audit Architecture
Edition	In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED

Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Memory & Context authority	NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED
Governance authority	NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED
Security authority	NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED
Prior baseline	NCIE-010 Evidence, Provenance & Audit Architecture, Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture

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Mandate, Authority Chain & Architecture Boundary

Establish NCIE-010 as the Evidence, Provenance & Audit architecture owner within the approved NCIE suite. NCIE-010 engineers the lifecycle and accountability layer around Evidence; it does not create canonical meaning, institutional authority or security controls of its own.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Master Controlled Register v2.0; NCIE-002 evidence/provenance-first doctrine; NCIE-003 Ch.17/25; NCIE-006/007 handover to NCIE-010.

1.0 Salient Things to Note

NCIE-010 owns Evidence integrity, lineage, provenance, accountability, audit and defensibility, but does not redefine NCIE-003 canonical entities, NCIE-008 governance authority, NCIE-009 security enforcement or NCIE-014 physical storage — Table 1 fixes this boundary so no later chapter can drift into owning what an upstream document already owns.

1.1 Mandate and Scope

NCIE-010 defines how source material becomes registered Evidence, how its integrity and custody are tracked, how it relates to other Evidence and to Findings/Decisions, and how enterprise provenance and audit records are structured — across every chapter and Agent/Tool interaction defined upstream.

1.2 Architecture Boundary — What NCIE-010 Does Not Own

NCIE-010 does not own canonical entity meaning, institutional governance, security enforcement, Memory semantics or physical storage. It owns the Evidence lifecycle and enterprise provenance/audit engineering layered across all of them.

Table 1. NCIE-010 architecture boundary.

Concern	Owned By
Product requirements/scope	NCIE-001
System architecture	NCIE-002
Canonical data model, Evidence Object meaning	NCIE-003
Engineering baseline	NCIE-004
ARGUS behavioral authority	NCIE-005
Memory & Context architecture	NCIE-006
AI/Agent/Model orchestration	NCIE-007
Institutional governance, RACI, risk acceptance	NCIE-008
Security, IAM, Zero-Trust enforcement	NCIE-009
Evidence lifecycle, chain of custody, provenance, audit architecture	NCIE-010 (this document)
Detailed interfaces, physical storage, operations, detailed testing	NCIE-011/014/015/016 (downstream)

1.3 Interfaces to Upstream Authority

Every consequential Evidence/audit control in this document is expressed as an enforcement of an NCIE-008 decision-right class and an NCIE-009 security control, never as a new institutional authority or security mechanism invented here. Where an accountable holder is not yet assigned upstream, this document states the control and defers the assignment to the applicable Human Review Register.

1.R RACI / Control Ownership

Table 2. Chapter 1 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Adopt/confirm NCIE-010 architecture	Accountable Authority Class (NCIE-008, to be assigned)	Evidence Architecture Owner (to be assigned)	NCIE-008/009 chapter owners	All institutional role holders

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-1-1: Confirm NCIE-010 acceptance authority and resolve any unresolved overlap with audit/assurance responsibilities already defined in NCIE-008/009.

Traceability: Master Controlled Register v2.0; NCIE-002 Ch.4; NCIE-003 Ch.17/25; NCIE-006/007 handover.

Evidence & Provenance Doctrine

Lock the epistemic and architectural separations governing Evidence, provenance, audit and institutional decisions, so provenance can support defensibility without ever becoming truth or authority on its own.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 evidence/provenance-first requirements; NCIE-003 semantic separations; NCIE-005/007 Human-Primary invariants.

2.0 Salient Things to Note

SOURCE DATA ≠ EVIDENCE; AI OUTPUT ≠ EVIDENCE; Finding ≠ Decision; provenance supports defensibility but does not itself create truth or authority — every chapter below is required to restate the relevant piece of this doctrine wherever its subject matter could otherwise be misread as creating Evidence, a Finding, or institutional truth by default.

2.1 Evidence, Provenance and Audit Doctrine

Evidence is source material that has been deliberately registered and qualified, not merely any data that exists in the system (SOURCE DATA ≠ EVIDENCE). Provenance records where something came from and how it was handled; it does not itself establish that the content is true or authoritative. Audit records accountable actions and decisions; it is a distinct record class from both Evidence and provenance.

2.2 Anti-Fabrication Boundary

This document does not invent evidentiary admissibility rules, institutional truth standards, or authority that provenance/audit records do not themselves carry. AI output remains AI-derived until an independent governed process establishes it as Evidence (AI / AGENT OUTPUT ≠ EVIDENCE); a VPF PASS is a content-quality disposition, not Evidence (VPF PASS ≠ EVIDENCE); and Evidence support for a hypothesis is not itself an institutional Finding or Decision (EVIDENCE SUPPORT ≠ INSTITUTIONAL DECISION; FINDING ≠ DECISION).

2.3 Defensibility Principle

Defensibility — the ability to explain and substantiate a Finding or Decision after the fact — is built from the combination of Evidence identity, custody, integrity, lineage and audit trail defined across this document; no single record class is sufficient alone, and none may substitute for institutional review under NCIE-008.

2.R RACI / Control Ownership

Table 3. Chapter 2 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Maintain/interpret the Evidence/Provenance/Audit doctrine	Accountable Authority Class (to be assigned)	Evidence Architecture Owner (to be assigned)	Legal/Policy Function (NCIE-008, to be assigned)	All chapter owners

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR10-2-1: Approve any institution-specific evidence admissibility principles not fixed upstream; does not block the doctrine already stated above in the interim.

Traceability: NCIE-001 evidence/provenance-first requirements; NCIE-003 semantic separations; NCIE-005/007 Human-Primary invariants.

Evidence Object Model & Canonical Interfaces

Define how NCIE-010 operationalizes the canonical Evidence Object and Source Object without redefining NCIE-003's ownership of their meaning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.17 Evidence & Provenance and Ch.25 provenance model.

3.0 Salient Things to Note

NCIE-003 owns canonical entity meaning; NCIE-010 owns lifecycle, integrity and chain-of-custody engineering around those entities — every interface this chapter defines is additive to the canonical object, never a restatement or reinterpretation of what NCIE-003 already fixed.

3.1 Canonical Interface, Not Redefinition

NCIE-003 owns what an Evidence Object and a Source Object mean; NCIE-010 owns the lifecycle, integrity and chain-of-custody engineering built around those canonical entities — this chapter defines the interface, not a competing definition.

3.2 Engineered Interfaces Around the Canonical Object

NCIE-010 attaches to every Evidence Object: an integrity-metadata reference (Chapter 6), a version/status reference (Chapter 8), and relationship interfaces to other Evidence, Hypotheses, Findings and Decisions (Chapters 10, 17). None of these interfaces alters the canonical Evidence Object's own identity or NCIE-003-defined meaning.

3.R RACI / Control Ownership

Table 4. Chapter 3 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve any Evidence-class extension requiring canonical amendment	Accountable Authority Class (to be assigned)	Evidence Architecture Owner (to be assigned)	NCIE-003 canonical data owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-3-1: Confirm any evidence-class extensions that require a canonical-data amendment (routed to NCIE-003) rather than NCIE-010-only treatment.

Traceability: NCIE-003 Ch.17 Evidence & Provenance, Ch.25 provenance model.

Source Registration & Evidence Qualification

Define the controlled, deliberate transition from source material or reference to a registered Evidence Object.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Evidence lifecycle step 01; NCIE-001 source-to-decision boundaries.

4.0 Salient Things to Note

Registration is deliberate and governed; merely ingesting, caching or retrieving source material does not make it Evidence — Table 5 fixes a provenance and audit obligation at every lifecycle stage so this boundary cannot be silently crossed at a later stage without a recorded event.

4.1 Registration Is Deliberate, Not Automatic

Merely ingesting, caching or retrieving source material does not make it Evidence. Source registration requires an identified source, an acquisition context, a registration actor or process, and an explicit qualification decision before Evidence status attaches.

4.2 Rejection and Quarantine

Source material that fails qualification is rejected or quarantined with a recorded reason; quarantined material remains traceable but is not treated as Evidence until qualification is satisfied or an accountable holder overrides quarantine through a recorded decision.

4.3 Evidence Lifecycle Matrix (directive item 50)

The logical lifecycle below traces every stage from source registration through disposition, showing the provenance and audit obligation each stage carries and the chapter that engineers it.

Table 5. Evidence Lifecycle Matrix.

Lifecycle Stage	Provenance Obligation	Audit Obligation	Owning Chapter
Source registration	Source identity, acquisition context, registration actor	Audit event: registration decision	Ch.4
Qualification	Qualification criteria applied, rejection/quarantine reason if applicable	Audit event: qualification outcome	Ch.4
Acquisition/capture	Channel, time, actor/process, method, environment	Audit event: acquisition completed	Ch.5
Integrity establishment	Content digest reference, integrity event	Audit event: integrity check performed	Ch.6
Custody (access/transfer)	Handler identity, purpose, timestamps	Audit event: custody transfer/access	Ch.7
Versioning/correction	Version lineage, correction reason, effective time	Audit event: correction/supersession recorded	Ch.8
Invalidation/withdrawal	Invalidation state, reason, downstream-impact trigger	Audit event: invalidation/withdrawal recorded	Ch.9
Transformation (derived Evidence)	Input/output Evidence, tool/process, parameters	Audit event: transformation recorded	Ch.13

Retention/disposition	Retention class reference, tombstone/history	Audit event: disposition decision	Ch.24
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4.R RACI / Control Ownership

Table 6. Chapter 4 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve Evidence qualification authority/criteria	Accountable Authority Class (to be assigned)	Source Registration Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-4-1: Approve Evidence qualification authorities and criteria not already established upstream.

Traceability: NCIE-003 Evidence lifecycle step 01; NCIE-001 source-to-decision boundaries.

Evidence Acquisition & Capture Provenance

Define the provenance required when Evidence is acquired from systems, humans, documents, tools, sensors or external sources.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 evidence acquisition workflows; NCIE-003 provenance requirements; NCIE-004/011 acquisition interfaces.

5.0 Salient Things to Note

Acquisition provenance must distinguish source, acquisition path, time, actor/process and transformation; later use must not obscure origin — every acquisition record persists unmodified alongside the Evidence Object regardless of how many transformations follow.

5.1 Acquisition Provenance Elements

Every acquisition event records source channel, timestamps, actor or service identity, capture method, environment/context, and the acquisition result. Later use of the Evidence must not obscure this origin — the acquisition record remains attached for the Evidence Object's full lifecycle.

5.2 Heightened Attestation Classes

Some acquisition classes (e.g., acquisition from an unverified external source, or acquisition under adversarial conditions) may warrant heightened human attestation or independent corroboration before qualification; which classes require this is a Human Review decision, not a default this document sets unilaterally.

5.R RACI / Control Ownership

Table 7. Chapter 5 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve heightened- attestation acquisition classes	Accountable Authority Class (to be assigned)	Acquisition Provenance Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-5-1: Confirm which acquisition classes require heightened human attestation or independent corroboration.

Traceability: NCIE-001 evidence acquisition workflows; NCIE-003 provenance requirements; NCIE-004/011 acquisition interfaces.

Integrity Metadata, Hashing & Tamper Evidence

Define the integrity metadata needed to detect alteration and support defensibility, without prescribing specific hash or cryptographic algorithms.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.17 lifecycle/integrity metadata; NCIE-009 artifact/integrity security.

6.0 Salient Things to Note

Hash/integrity metadata can show sameness/change, not truthfulness or substantive reliability of the content — this chapter never allows a passed integrity check to be cited as evidence of the content's accuracy, only of its unaltered state since a recorded point.

6.1 Integrity Shows Sameness, Not Truth

HASH / INTEGRITY METADATA ≠ TRUTH OF CONTENT. Integrity metadata (a content digest reference, an integrity event, verification results) can show whether content is unchanged since registration or capture; it cannot establish that the content is accurate, complete or reliable.

6.2 Mismatch Handling and Re-Verification

A detected integrity mismatch triggers tamper-indication handling: the Evidence is flagged, the mismatch is recorded with time and detecting actor/process, and re-verification or escalation follows rather than silent continued use. The specific hash/cryptographic algorithm is an engineering choice made where NCIE-004/009 fix it, not invented here.

6.R RACI / Control Ownership

Table 8. Chapter 6 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve integrity assurance requirements by Evidence class	Accountable Authority Class (to be assigned)	Integrity Assurance Owner (to be assigned)	NCIE-009 artifact-security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-6-1: Approve integrity assurance requirements by Evidence class without inventing specific algorithms prematurely.

Traceability: NCIE-003 Ch.17 lifecycle/integrity metadata; NCIE-009 artifact/integrity security.

Chain of Custody Architecture

Define custody and handling provenance from registration through access, transfer, transformation, correction, invalidation and disposition.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.17 §17.6; NCIE-006/007 explicitly hand chain-of-custody engineering to NCIE-010.

7.0 Salient Things to Note

Chain of custody records who/what handled Evidence and how; it does not establish content ownership or substantive truth — Table 9 fixes the same core fields (handler, purpose, integrity check, state transition) across every custody-event class so no event type can skip an integrity check or leave the handler unattributed.

7.1 Custody Records Handling, Not Ownership

CHAIN OF CUSTODY ≠ CONTENT OWNERSHIP. A custody event records who or what handled Evidence, when, why and with what integrity check — it establishes accountable handling, not who owns or authored the underlying content.

7.2 Dual-Control and Formal Attestation

Which Evidence classes require formal custody attestation or dual-control handling (e.g., two independent handlers confirming a transfer) is a Human Review decision; this document defines the custody-event structure that any such requirement would be recorded against.

7.3 Chain-of-Custody Matrix (directive item 51)

Every custody-event class below carries the same core fields, without inventing institutional actors beyond the abstract handler role.

Table 9. Chain-of-Custody Matrix.

Custody Event	Handler Identity	Purpose	Transfer/Access Type	Integrity Check	State Transition
Registration	Registering actor/process	Initial custody establishment	N/A (origin)	Initial integrity metadata recorded	Unregistered → Registered
Access	Accessing principal (NCIE-009 authenticated)	Review/analysis	Read access	Integrity verification on read (optional per policy)	No state change
Transfer	Sending and receiving handler	Custody handoff	Transfer	Integrity verification at transfer	Custody holder updated
Transformation	Transforming actor/process/Tool	Derived-Evidence creation (Ch.13)	Read + derived write	Integrity metadata on output artifact	New derived Evidence created; source unchanged
Correction	Correcting actor (governed, Ch.8)	Version correction	Write (new version)	Integrity metadata on new version	Previous → Current version
Disposition	Disposition-approving actor (NCIE-008)	Retention/disposition (Ch.24)	N/A (terminal)	Final integrity/tombstone record	Active → Disposed (history)

retained)

7.R RACI / Control Ownership

Table 10. Chapter 7 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve custody-attestation/dual-control classes	Accountable Authority Class (to be assigned)	Chain-of-Custody Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-7-1: Confirm which Evidence classes require formal custody attestation or dual-control.

Traceability: NCIE-003 Ch. 17 §17.6; NCIE-006/007 chain-of-custody handover to NCIE-010.

Evidence Versioning, Correction & Supersession

Define controlled handling of corrected or superseded Evidence without rewriting history.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Evidence lifecycle; NCIE-006 dependency/revalidation semantics.

8.0 Salient Things to Note

Correction creates a new governed version/state; original Evidence history remains reconstructable — this chapter's previous/current relationship model is the structural enforcement of CORRECTION ≠ HISTORICAL ERASURE, not merely a stated principle.

8.1 Correction Creates a New Version

CORRECTION ≠ HISTORICAL ERASURE. A correction produces a new governed version with its own effective time and a recorded reason; the original Evidence version remains reconstructable, linked via a previous/current relationship rather than overwritten.

8.2 Notification and Impact

Where a correction affects downstream Findings, Decisions or dependent Evidence, a notification/impact interface signals affected consumers; whether corrected versions require human validation before becoming current is Evidence-class-dependent and left to Human Review.

8.R RACI / Control Ownership

Table 11. Chapter 8 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve correction authority/validation requirements	Accountable Authority Class (to be assigned)	Evidence Versioning Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-8-1: Approve correction authority and confirm whether any Evidence classes require human validation before a corrected version becomes current.

Traceability: NCIE-003 Evidence lifecycle; NCIE-006 dependency/revalidation semantics.

Evidence Invalidation, Withdrawal & Reliability State

Define how Evidence becomes invalidated, withdrawn, restricted or reliability-qualified while preserving historical traceability.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Evidence lifecycle step 06; NCIE-006/007 dependency-impact rules.

9.0 Salient Things to Note

Invalidation does not delete the historical record; historical records remain available according to governance/security rules while current use reflects the changed state — this chapter treats invalidation as a state transition to record, never as a basis for removing the prior record.

9.1 Invalidation Is Not Deletion

INVALIDATION ≠ DELETION OF HISTORICAL RECORD. Invalidated, withdrawn or restricted Evidence remains available historically according to governance/security rules; current use reflects the changed state (e.g., excluded from active analysis) without erasing the record that it once had a different state.

9.2 Appeal and Downstream Impact

Every invalidation/withdrawal carries a reason, an effective time, and a downstream-impact trigger notifying dependent Findings/Decisions. Invalidation/withdrawal authority and any appeal or review process is a Human Review decision.

9.R RACI / Control Ownership

Table 12. Chapter 9 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve invalidation/withdrawal authority and appeal process	Accountable Authority Class (to be assigned)	Evidence Reliability Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-9-1: Confirm invalidation/withdrawal authority and the appeal/review process.

Traceability: NCIE-003 Evidence lifecycle step 06; NCIE-006/007 dependency-impact rules.

Evidence Relationships, Support & Contradiction

Define governed relationships among Evidence and between Evidence, Hypotheses, Findings and Decisions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.17 relationship model and contradiction-preservation principle.

10.0 Salient Things to Note

Contradicting Evidence must be preserved; support/contradiction relationships do not themselves resolve the dispute or create a Finding — Table 13's semantic-caution column makes this explicit for every relationship type so none can be read as conferring resolution authority.

10.1 Relationships Do Not Resolve Disputes

A support or contradiction relationship between two Evidence items records that the relationship exists; it does not itself resolve the dispute, and it does not create a Finding. Contradicting Evidence must be preserved, never silently dropped because it disagrees with other Evidence.

10.2 Relationship Provenance and Qualification

Every relationship carries its own provenance (who/what declared it, when, on what basis) and may carry a confidence or qualification value; relationship vocabulary extensions and rules for declaring independent corroboration are Human Review decisions.

10.3 Evidence Relationship Matrix (directive item 52)

The controlled relationship vocabulary below fixes allowed endpoints and semantic cautions for each relationship type.

Table 13. Evidence Relationship Matrix.

Relationship	Allowed Endpoints	Semantic Caution
Supports	Evidence → Hypothesis/Evidence	Does not itself create a Finding or resolve a dispute
Contradicts	Evidence → Evidence/Hypothesis	Must be preserved, never silently removed or summarized away
Corroborates	Evidence → Evidence	Independent-source corroboration rules are Human Review, not inferred by default
Duplicates	Evidence → Evidence	Marking as duplicate does not invalidate either copy's own custody/integrity record
Derives-from	Derived Evidence → Source Evidence	Preserves lineage (Ch.13); does not erase source identity
Related-to	Evidence ↔ Evidence/Hypothesis/Finding/Decision	Generic association only — carries no support/contradiction weight

10.R RACI / Control Ownership

Table 14. Chapter 10 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve relationship vocabulary	Accountable Authority Class (to	Evidence Relationship	Reviewer	Assurance/Audit

extensions/corroboration rules	be assigned)	Owner (to be assigned)	(NCIE-008)
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Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR10-10-1: Approve relationship vocabulary extensions and any rules for declaring independent corroboration; does not block the six relationship types already fixed in Table 13.

Traceability: NCIE-003 Ch. 17 relationship model and contradiction-preservation principle.

Contradiction Preservation & Evidence Gaps

Ensure unresolved contradictions and missing evidence remain visible through analysis and decision support.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/005 evidence-gap behavior; NCIE-006 contradiction preservation; NCIE-007 grounding package.

11.0 Salient Things to Note

Evidence gaps must survive summaries, AI transformations and Context compression — this chapter's contradiction-set and gap-record structures are the mechanism that makes CONTRADICTION ≠ ERROR TO BE SILENTLY REMOVED enforceable rather than aspirational.

11.1 Contradiction Is Not an Error to Remove

CONTRADICTION ≠ ERROR TO BE SILENTLY REMOVED. A contradiction set (two or more Evidence items that conflict) and an evidence gap (a known absence of Evidence needed for a conclusion) are both preserved as explicit records, not resolved by picking one side or omitting the gap.

11.2 Survival Through Summaries and Compression

Contradictions and gaps must survive AI-generated summaries, transformations and Context compression (NCIE-006/007) — a summary that drops a contradiction because it complicates the narrative is a defect, not an acceptable simplification. Resolution requires an explicit resolution event, not silent disappearance.

11.R RACI / Control Ownership

Table 15. Chapter 11 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve contradiction-resolution governance	Accountable Authority Class (to be assigned)	Contradiction/Gap Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-11-1: Confirm the governance for declaring a contradiction resolved versus merely superseded.

Traceability: NCIE-001/005 evidence-gap behavior; NCIE-006 contradiction preservation; NCIE-007 grounding package.

Annotation, Commentary & Expert Interpretation

Separate annotations, commentary and expert interpretations from the underlying Evidence they discuss.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Evidence lifecycle step 04; NCIE-005 ARGUS provenance; NCIE-007 AI output boundaries.

12.0 Salient Things to Note

Human or ARGUS annotation remains separately attributable and versioned from the Evidence it discusses — this chapter's annotation object is structurally incapable of modifying the underlying Evidence Object it references.

12.1 Annotation Never Alters Source Evidence

ANNOTATION ≠ ALTERATION OF SOURCE EVIDENCE. An annotation (human or ARGUS-originated) is a separately attributable, separately versioned layer referencing the Evidence — it never modifies the Evidence Object's own identity, content or integrity metadata.

12.2 Authoritative Expert Opinion

Every annotation carries author/origin, timestamp, scope, interpretation type and a confidence/qualification value, with full edit history. Which annotation classes may be treated as authoritative expert opinion, and under what review, is a Human Review decision.

12.R RACI / Control Ownership

Table 16. Chapter 12 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve authoritative expert-annotation classes	Accountable Authority Class (to be assigned)	Annotation Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR10-12-1: Confirm which annotation classes may be considered authoritative expert opinion and under what review; does not block ordinary non-authoritative annotation in the interim.

Traceability: NCIE-003 Evidence lifecycle step 04; NCIE-005 ARGUS provenance; NCIE-007 AI output boundaries.

Derived Evidence & Transformation Lineage

Define lineage for transformations such as extraction, normalization, redaction, transcription, aggregation or forensic derivation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002/003 lineage requirements; NCIE-004 processing pipelines; NCIE-009 DLP/redaction/security.

13.0 Salient Things to Note

Derived artifacts must retain lineage to source Evidence and transformations; transformation does not erase original source identity — every transformation event in this chapter's model is additive to the lineage graph, never a replacement of the source record.

13.1 Transformation Preserves Source Identity

Derived artifacts retain lineage to their source Evidence and to the transformation(s) that produced them; transformation never erases the original source's identity. Each transformation event records input Evidence, output artifact, the tool/process and version used, parameters or configuration reference, integrity metadata on the output, and known quality/limitations.

13.2 Which Transformations Yield New Evidence

Not every transformation output is itself Evidence — some remain non-Evidence analytical artifacts. Which transformation classes may yield new Evidence versus remain analytical artifacts only is a Human Review decision, applied consistently through the lineage record either way.

13.R RACI / Control Ownership

Table 17. Chapter 13 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve transformation-to-Evidence classification rules	Accountable Authority Class (to be assigned)	Derived-Evidence Owner (to be assigned)	NCIE-009 DLP/redaction owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-13-1: Approve which transformations may yield new Evidence versus remain non-Evidence analytical artifacts.

Traceability: NCIE-002/003 lineage requirements; NCIE-004 processing pipelines; NCIE-009 DLP/redaction/security.

AI / Agent / ARGUS Provenance Boundary

Define provenance for AI-derived summaries, analyses, hypotheses and transformations that reference Evidence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005/006/007 AI provenance and Evidence boundaries; NCIE-003 Validated AI Result model.

14.0 Salient Things to Note

AI/Agent/ARGUS output remains AI-derived unless an independent governed process establishes a legitimate Evidence object; VPF PASS ≠ EVIDENCE — this chapter's output-classification field makes this status explicit and machine-checkable on every AI-derived artifact, not merely documented in prose.

14.1 AI Output Remains AI-Derived

AI / Agent / ARGUS output remains AI-derived unless an independent governed process establishes it as a legitimate Evidence object — VPF PASS ≠ EVIDENCE, and no AI output becomes Evidence merely by being referenced, summarized or cited within the system.

14.2 Provenance Fields for AI-Derived Content

Every AI-derived artifact that references Evidence records its AI origin, the Agent Run/model route, the Context/Evidence references it drew on, its VPF state, any human modification applied afterward, and an explicit output classification (AI-derived, human-reviewed, promoted-to-Evidence-pending-review, etc.). Which AI-assisted acquisition or transformation workflows may ever produce Evidence, and what human controls they require, is a Human Review decision.

14.R RACI / Control Ownership

Table 18. Chapter 14 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve AI-assisted acquisition/transformation Evidence controls	Accountable Authority Class (to be assigned)	AI Provenance Owner (to be assigned)	NCIE-007 orchestration owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-14-1: Confirm whether any AI-assisted acquisition/transformation workflows can produce Evidence and what human controls they require.

Traceability: NCIE-005/006/007 AI provenance and Evidence boundaries; NCIE-003 Validated AI Result model.

Human Contribution & Attribution Provenance

Define attribution for human observations, annotations, assessments, approvals and corrections linked to Evidence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/005 Human-Primary provenance; NCIE-008 governance roles.

15.0 Salient Things to Note

Human-origin does not automatically make a statement Evidence or a Finding; provenance must identify the role/type of contribution — this chapter's contribution-type field is mandatory on every human-attributed record precisely to prevent that automatic elevation.

15.1 Human Origin Alone Is Not Evidence or a Finding

A statement is not automatically Evidence or a Finding merely because a human made it. Attribution provenance identifies the human principal, their governance role (if any), the contribution type (observation, annotation, assessment, approval, correction), the time, the source/Evidence referenced, and the review/approval state.

15.2 Specialized Contribution Classes

Whether institution-specific contribution classes (e.g., expert witness, analyst, validator) require distinct provenance treatment is a Human Review decision; this document defines the generic attribution structure such a class would use.

15.R RACI / Control Ownership

Table 19. Chapter 15 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve specialized human-contribution classes	Accountable Authority Class (to be assigned)	Attribution Provenance Owner (to be assigned)	NCIE-008 role-taxonomy owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR10-15-1: Confirm expert-witness/analyst/validator contribution classes if institutionally required; does not block the generic attribution structure already defined in the interim.

Traceability: NCIE-003/005 Human-Primary provenance; NCIE-008 governance roles.

Temporal Semantics & Decision-Time Evidence State

Preserve the Evidence state that existed at historical decision cutoffs, distinct from current state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 temporal semantics; NCIE-006 Decision-Time architecture; NCIE-007 model/Tool historical state.

16.0 Salient Things to Note

Later corrections, Evidence, invalidations or approvals must not silently contaminate the historical view — Table 20 fixes exactly what historical state each artifact class needs and how an unavailable state is flagged rather than backfilled with current data.

16.1 Reconstruction Is Not Rewrite

DECISION-TIME RECONSTRUCTION ≠ CURRENT-STATE REWRITE. Reconstructing what was known at a past decision cutoff uses the historical version, observed time, effective time and registration time that actually existed then — later corrections, new Evidence, invalidations or approvals must not silently contaminate that historical view.

16.2 Temporal Fields

Every Evidence Object carries observed time (when the underlying fact occurred), effective time (when a version became authoritative), and registration time (when it entered the system) — distinct fields that together let a reconstruction query select exactly the state available at any chosen cutoff.

16.3 Decision-Time Reconstruction Matrix (directive item 54)

The table below identifies the historical state needed to reconstruct each artifact class at a cutoff, and how an unavailable state is handled.

Table 20. Decision-Time Reconstruction Matrix.

Artifact Class	Historical State Needed	Unavailable-State Handling
Evidence	Version current as of cutoff (effective time ≤ cutoff)	Explicitly flagged as NOT AVAILABLE AT CUTOFF, never substituted with current version
Transformation/derived artifact	Transformation version and parameters as of cutoff	Flagged unavailable; not silently regenerated with current parameters
Provenance record	Provenance graph edges existing as of cutoff	Later-added edges excluded from the reconstructed view
Audit record	Audit events with time ≤ cutoff	Later audit events excluded
Tool outcome (NCIE-007)	Outcome/receipt as recorded at cutoff, including OUTCOME UNKNOWN if still unresolved then	Later reconciliation not backdated into the historical view
AI artifact	Model route, Context and VPF state as of cutoff (NCIE-007 Ch.9)	Later revalidation excluded from historical view (REVALIDATION ≠ HISTORICAL REWRITE)
Finding/Decision	Evidentiary basis as of the Decision's own effective time	Later supporting or contradicting Evidence excluded from what the Decision-maker actually had

16.R RACI / Control Ownership

Table 21. Chapter 16 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve retention horizon/reconstruction authority	Accountable Authority Class (to be assigned)	Temporal Semantics Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-16-1: Confirm retention horizon and authority for formal historical reconstruction where not fixed upstream.

Traceability: NCIE-003 temporal semantics; NCIE-006 Decision-Time architecture; NCIE-007 model/Tool historical state.

Evidence-to-Hypothesis / Finding / Decision Traceability

Define traceability from Evidence through hypotheses and formal Findings/Decisions without collapsing the distinct states.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 source-to-decision boundary; NCIE-003 Ch.16/17; NCIE-005 Human-Primary boundaries.

17.0 Salient Things to Note

Evidence can support/contradict reasoning; it does not itself become a Finding or Decision — every institutional Finding/Decision in this chapter's traceability model remains separately attributable to the NCIE-008 authority that issued it.

17.1 Distinct States, Traceable Chain

Evidence can support or contradict reasoning; it does not itself become a Finding or Decision. The traceability chain records the Evidence basis, any hypothesis link, the Finding's support set, the Decision's support set, and any dissent or counterevidence — each state remains separately attributable to its own accountable authority (NCIE-008 Ch.18).

17.2 Minimum Evidentiary Package

The minimum evidentiary package and authority required to validate a Finding, where not already resolved by NCIE-008, is a Human Review decision addressed jointly with Chapter 27's defensibility package.

17.R RACI / Control Ownership

Table 22. Chapter 17 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve minimum evidentiary package for Finding validation	Accountable Authority Class (to be assigned)	Traceability Owner (to be assigned)	NCIE-008 Ch.18 Finding/Decision owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-17-1: Confirm the minimum evidentiary package/authority for validating Findings where unresolved.

Traceability: NCIE-001 source-to-decision boundary; NCIE-003 Ch.16/17; NCIE-005 Human-Primary boundaries.

Provenance Graph & Lineage Architecture

Define enterprise lineage/provenance relationships across sources, Evidence, transformations, AI artifacts, Findings and Decisions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.17/19/25; NCIE-002 graph/provenance architecture.

18.0 Salient Things to Note

Graph connectivity cannot be represented as causal proof without governed analytical support — this chapter's provenance queries are scoped to answer 'what is connected to what and how,' never 'what caused what.'

18.1 Lineage Is Not Causality

LINEAGE ≠ CAUSALITY. Graph connectivity between nodes (Evidence, transformations, AI artifacts, Findings, Decisions) shows that a relationship was recorded; it cannot be represented as causal proof of one thing producing or explaining another without governed analytical support external to the graph itself.

18.2 Graph Structure

The provenance graph's nodes and edges carry version references, actor/process references and temporal edges, and support provenance queries (e.g., 'what led to this Finding'); institution-specific extensions to the relationship vocabulary beyond the canonical baseline are a Human Review decision.

18.R RACI / Control Ownership

Table 23. Chapter 18 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve provenance relationship vocabulary extensions	Accountable Authority Class (to be assigned)	Provenance Graph Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR10-18-1: Approve any institution-specific provenance relationship vocabulary beyond the canonical baseline; does not block the baseline graph model already defined.

Traceability: NCIE-003 Ch.17/19/25; NCIE-002 graph/provenance architecture.

Audit Event Architecture

Define authoritative audit-event semantics distinct from provenance and telemetry.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Audit Event model; NCIE-006/007/009 telemetry-audit-provenance boundaries.

19.0 Salient Things to Note

Audit exists to establish accountable actions/decisions, not merely operational observability — Table 24 makes the three-way distinction explicit by purpose, object, authority, retention and security sensitivity so no later chapter can conflate them.

19.1 Audit Is Accountability, Not Observability

AUDIT EVENT ≠ PROVENANCE RECORD, and TELEMETRY ≠ AUDIT ≠ PROVENANCE. Audit exists to establish accountable actions and decisions — who did what, to what, with what authorization result — not merely to observe system operation.

19.2 Audit Event Fields

Every audit event records actor, action, target, decision/outcome, reason/purpose, time, authorization result, and a correlation/reference to related events. Which audit-event classes require heightened assurance or independent review is a Human Review decision.

19.3 Provenance versus Audit Matrix (directive item 53)

Provenance, audit and telemetry are explicitly compared below so they cannot be conflated in downstream design or in this specification's own later chapters.

Table 24. Provenance versus Audit versus Telemetry Matrix.

Dimension	Provenance	Audit	Telemetry
Purpose	Show origin, lineage and handling of Evidence/artifacts	Establish accountable actions and decisions	Operational observability of system behavior
Object	Evidence, Sources, transformations, derived artifacts	Actor actions, decisions, access requests	System/service performance and health events
Authority	Never itself an authority to disclose or decide (PROVENANCE ≠ AUTHORITY TO DISCLOSE)	Never itself a grant of access or approval	Never itself audit or provenance evidence
Retention interface	NCIE-010 Ch.24 logical retention (downstream NCIE-014)	NCIE-010 Ch.24, heightened for accountability records	NCIE-007 Ch.31 operational retention, separate from Ch.24
Security sensitivity	May reveal sensitive lineage/identities (Ch.23)	May reveal sensitive institutional deliberation (Ch.21)	Generally lower sensitivity, still subject to NCIE-009 controls
Decision-Time use	Reconstructable per Ch.16 historical state rules	Reconstructable per Ch.16 historical state rules	Not authoritative for Decision-Time reconstruction

19.R RACI / Control Ownership

Table 25. Chapter 19 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve heightened- assurance audit- event classes	Accountable Authority Class (to be assigned)	Audit Architecture Owner (to be assigned)	NCIE-008 Ch.23 assurance owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-19-1: Confirm audit-event classes requiring heightened assurance or independent review.

Traceability: NCIE-003 Audit Event model; NCIE-006/007/009 telemetry-audit-provenance boundaries.

Authorization & Access Audit

Define audit requirements for evidence access, disclosure, export, modification and denied attempts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 authorization audit; NCIE-009 IAM/Zero Trust; NCIE-008 governance.

20.0 Salient Things to Note

Audit records current authorization decisions and actions but does not itself grant disclosure authority — this chapter never allows a complete audit trail to be cited as a substitute for a live NCIE-009 authorization check.

20.1 Audit Records the Decision, Not the Authority

Audit records current authorization decisions and actions but does not itself grant disclosure authority — access remains governed by NCIE-009 IAM/Zero Trust controls at the moment of the request, independent of what the audit trail later shows.

20.2 Denials Matter As Much As Permits

Every access request — permit or deny — is recorded with principal, purpose, object/field scope, and whether it involved disclosure/export or privileged access. Denied attempts are not discarded; they are often the more security-relevant record.

20.R RACI / Control Ownership

Table 26. Chapter 20 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve heightened-audit access classes/reviewers	Accountable Authority Class (to be assigned)	Access Audit Owner (to be assigned)	NCIE-009 IAM owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR10-20-1: Confirm heightened-audit access classes and who may review sensitive audit records.

Traceability: NCIE-001 authorization audit; NCIE-009 IAM/Zero Trust; NCIE-008 governance.

Governance Decision & Approval Audit

Define auditability for institutional approvals, exceptions, Human Review outcomes and governance decisions affecting Evidence/provenance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008 decision-right/RACI/approval evidence; NCIE-009 governance-security interface.

21.0 Salient Things to Note

Governance Decision records must remain distinct from AI/VPF states and technical execution receipts — this chapter's audit record never accepts a Gate pass, VPF PASS or execution receipt as a substitute for the NCIE-008 decision record it is meant to audit.

21.1 Governance Decisions Stay Distinct

A Governance Decision record (approver/role, target/scope/version, conditions, effective time, revocation/supersession, evidentiary basis) remains distinct from AI/VPF states and from technical execution receipts — the same separation NCIE-008 Ch.10 and Ch.18 already require, applied here to Evidence/provenance-affecting decisions specifically.

21.2 Sensitive Deliberation

Governance deliberation (exception requests, Human Review debate) may itself be sensitive; audit retention and access requirements for such records are a Human Review decision.

21.R RACI / Control Ownership

Table 27. Chapter 21 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve sensitive-governance-deliberation audit handling	Accountable Authority Class (to be assigned)	Governance Audit Owner (to be assigned)	NCIE-008 Ch.10/18 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR10-21-1: Confirm audit retention/access requirements for sensitive governance deliberation; does not block ordinary governance-decision audit capture in the interim.

Traceability: NCIE-008 decision-right/RACI/approval evidence; NCIE-009 governance-security interface.

Tool, Workflow & External-System Provenance

Define provenance and audit links for Tool calls, workflows, connectors and external-system actions that acquire or affect Evidence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Tool Gateway; NCIE-007 Tool execution receipts; NCIE-009 connector security.

22.0 Salient Things to Note

Tool proposal ≠ execution; submission ≠ confirmed outcome. OUTCOME UNKNOWN must remain visible until reconciled — this chapter never permits an unresolved outcome to be silently treated as successful just because time has passed.

22.1 Proposal Is Not Execution

A Tool proposal or submission is not a confirmed outcome — each invocation records the actor/effective principal, target, request, receipt, external identifier and outcome, with reconciliation before the Evidence impact (if any) is treated as settled.

22.2 OUTCOME UNKNOWN Stays Visible

Where an external action's outcome cannot be confirmed, it is recorded and displayed as OUTCOME UNKNOWN until reconciled — never silently assumed successful, and never quietly dropped from the record while unresolved. Which external actions require additional evidentiary receipts or dual confirmation is a Human Review decision.

22.R RACI / Control Ownership

Table 28. Chapter 22 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve dual-confirmation/receipt requirements for external actions	Accountable Authority Class (to be assigned)	Tool/Workflow Provenance Owner (to be assigned)	NCIE-009 connector-security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-22-1: Confirm which external actions require additional evidentiary receipts or dual confirmation.

Traceability: NCIE-004 Tool Gateway; NCIE-007 Tool execution receipts; NCIE-009 connector security.

Security, Privacy & Classification of Evidence/Audit Records

Apply NCIE-009 security constraints to Evidence, provenance and audit artifacts without redefining IAM.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 classification; NCIE-006 privacy; NCIE-009 Security/IAM/DLP; NCIE-008 disclosure governance.

23.0 Salient Things to Note

Audit/provenance may itself reveal sensitive identities, investigations or protected data — this chapter requires classification to inherit into provenance and audit records rather than treating them as a lower-sensitivity byproduct of the Evidence they describe.

23.1 Retrievability Is Not Disclosure Authority

PROVENANCE ≠ AUTHORITY TO DISCLOSE. Audit and provenance records may themselves reveal sensitive identities, investigations or protected data — their existence or retrievability never authorizes disclosure; current NCIE-009 classification, IAM, DLP and protected-identity controls apply to them exactly as to the underlying Evidence.

23.2 Inheritance and Masking

Classification inherits from the underlying Evidence/Source into its provenance and audit records; sensitive metadata (e.g., a protected identity appearing only in a custody log) is masked consistently with the same rule that would mask it in the Evidence itself.

23.R RACI / Control Ownership

Table 29. Chapter 23 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve special-handling classes for sensitive Evidence/audit records	Accountable Authority Class (to be assigned)	Evidence Security/Classification Owner (to be assigned)	NCIE-009 classification/DLP owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR10-23-1: Confirm special handling classes for highly sensitive Evidence/provenance/audit records.

Traceability: NCIE-003 classification; NCIE-006 privacy; NCIE-009 Security/IAM/DLP; NCIE-008 disclosure governance.

Retention, Legal Hold, Archival & Disposition Interface

Define logical retention, legal hold, archival and disposition interfaces without choosing physical storage or inventing periods.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 retention/legal hold; NCIE-008 governance; NCIE-014 physical storage downstream.

24.0 Salient Things to Note

Deletion/disposition must not falsify required history; legal hold/records policy overrides ordinary cleanup where applicable — this chapter's tombstone/history reference is the mechanism that keeps disposition truthful without preempting NCIE-014's physical storage choice.

24.1 Disposition Must Not Falsify History

Deletion or disposition of Evidence/provenance/audit records must not falsify required history — where a record is disposed, a tombstone/history reference preserves the fact that it existed and was disposed, consistent with governance and security rules; legal hold overrides ordinary disposition where applicable.

24.2 Logical Interface Only

This chapter defines a retention-class reference, legal-hold flag, archival state and disposition-approval interface — it does not select a physical storage system (NCIE-014) or invent retention periods; retention authorities and specific periods are Human Review decisions.

24.R RACI / Control Ownership

Table 30. Chapter 24 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve retention authorities/periods and legal-hold ownership	Accountable Authority Class (to be assigned)	Retention/Disposition Owner (to be assigned)	NCIE-014 physical-storage owner (downstream)	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR10-24-1: Confirm retention authorities/periods and legal-hold ownership where unresolved.

Traceability: NCIE-006 retention/legal hold; NCIE-008 governance; NCIE-014 physical storage downstream.

Evidence Search, Discovery & Retrieval Provenance

Define traceability for evidence discovery and search without making search indexes authoritative.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-006 search/vector boundaries; NCIE-009 retrieval security; NCIE-003 Evidence identity.

25.0 Salient Things to Note

Discovery must resolve to governed Evidence objects and current authorization; semantic similarity is not authority to disclose — a search index entry in this chapter is never treated as a substitute for checking the Evidence object's own current state.

25.1 Search Result Is Not Evidence State

SEARCH RESULT ≠ EVIDENCE STATE. A search index entry is a discovery aid, not the Evidence itself; every discovery must resolve to the governed Evidence object and its current authorization state before being relied upon. Semantic similarity is not authority to disclose (SEMANTIC RETRIEVAL ≠ AUTHORITY TO DISCLOSE, inherited from NCIE-008/009).

25.2 Query Provenance and Index Freshness

Each search records query provenance, result references, the authorization filtering applied, index freshness (how current the index is relative to Evidence state), source resolution, and a retrieval receipt. Discoverability rules for restricted or withdrawn Evidence, and for cross-investigation discovery, are Human Review decisions.

25.R RACI / Control Ownership

Table 31. Chapter 25 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve discoverability rules for restricted/withdrawn Evidence	Accountable Authority Class (to be assigned)	Search/Retrieval Provenance Owner (to be assigned)	NCIE-009 retrieval-security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-25-1: Confirm discoverability rules for restricted/withdrawn Evidence and cross-investigation discovery.

Traceability: NCIE-006 search/vector boundaries; NCIE-009 retrieval security; NCIE-003 Evidence identity.

Audit & Provenance Reconciliation / Recovery

Define reconciliation after outages, restore, delayed events or partial failure affecting audit or provenance records.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-004/006/009 recovery; NCIE-007 idempotency/reconciliation.

26.0 Salient Things to Note

Restored logs/indexes do not automatically become authoritative current state, and gaps must be explicitly identified — this chapter's restored-state qualification field is mandatory output of every reconciliation, not an optional footnote.

26.1 Recovery Is Not Reauthorization, and Gaps Are Explicit

RECOVERY ≠ REAUTHORIZATION, and restored logs or indexes do not automatically become authoritative current state. Reconciliation includes replay, duplicate suppression, missing-event detection, integrity verification, and explicit qualification of restored state — any remaining gap is identified explicitly, never silently closed.

26.2 Declaring Reconstruction Complete

Recovery provenance records what was reconciled, what remained missing, and when. The authority for declaring an audit/provenance reconstruction complete after a material gap is a Human Review decision — this document does not self-certify completeness.

26.R RACI / Control Ownership

Table 32. Chapter 26 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Declare audit/provenance reconstruction complete after a gap	Accountable Authority Class (to be assigned)	Reconciliation/Recovery Owner (to be assigned)	Assurance/Audit (NCIE-008)	Affected domain owners

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-26-1: Confirm the authority for declaring audit/provenance reconstruction complete after material gaps.

Traceability: NCIE-004/006/009 recovery; NCIE-007 idempotency/reconciliation.

Verification, Assurance & Defensibility Testing

Define the test and assurance classes needed to demonstrate Evidence/provenance/audit integrity without duplicating NCIE-016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 acceptance criteria; NCIE-008 assurance; NCIE-009 test-status discipline; NCIE-016 downstream.

27.0 Salient Things to Note

TEST SPECIFIED ≠ TEST EXECUTED. Defensibility requires demonstrable lineage, custody, contradiction preservation, temporal reconstruction and audit integrity — Table 33 specifies eight test classes and §27.3's defensibility package, with every status field marked to prevent specification from being mistaken for execution.

27.1 Test Classes

Eight test classes are specified: chain-of-custody, tamper detection, version-history integrity, contradiction preservation, Decision-Time reconstruction, audit completeness, authorization-audit, and recovery. Each traces to the chapter(s) whose controls it verifies.

Table 33. Evidence/provenance/audit test-class specification.

Test ID	Test Class	Verifies	Status
EVT-1	Chain-of-Custody Test	Ch.7: every custody event records handler, purpose and integrity check with no gap in the chain	TEST SPECIFIED
EVT-2	Tamper Detection Test	Ch.6: integrity mismatches are detected and trigger tamper-indication handling, not silent continued use	TEST SPECIFIED
EVT-3	Version-History Integrity Test	Ch.8-9: corrections/invalidations create new state without erasing the historical record	TEST SPECIFIED
EVT-4	Contradiction Preservation Test	Ch.10-11: contradictions/gaps survive AI summaries and Context compression	TEST SPECIFIED
EVT-5	Decision-Time Reconstruction Test	Ch.16: a historical cutoff query returns only state available at that cutoff	TEST SPECIFIED
EVT-6	Audit Completeness Test	Ch.19-21: every accountable action/decision produces a corresponding audit event	TEST SPECIFIED
EVT-7	Authorization-Audit Test	Ch.20: permits and denials are both recorded and audit access itself does not grant disclosure	TEST SPECIFIED
EVT-8	Recovery Test	Ch.26: reconciliation after	TEST SPECIFIED

outage/restore explicitly
flags gaps rather than
silently closing them

27.2 Test-Status Integrity

Specifying a test is not executing it (TEST SPECIFIED ≠ TEST EXECUTED). For every test above: IMPLEMENTATION EVIDENCE AVAILABLE = NO; TEST EXECUTED = NO; TEST PASSED = N/A; OPERATIONAL/HUMAN ACCEPTANCE = PENDING. Detailed test-case authoring and execution are downstream in NCIE-016.

27.3 Evidence Defensibility Package (directive item 55)

To defend a material Finding or Decision, the following logical package is required — without inventing institutional admissibility rules: (1) source identity and registration record; (2) all relevant Evidence versions with correction/supersession history; (3) the complete chain-of-custody record; (4) transformation/derivation lineage for any derived Evidence relied upon; (5) any known contradicting Evidence and unresolved gaps; (6) governance approvals (NCIE-008) supporting the Finding/Decision; and (7) the applicable audit references (Ch.19-21). None of these seven elements alone is sufficient; assembling the package does not itself constitute institutional review or approval, which remains NCIE-008's authority.

27.R RACI / Control Ownership

Table 34. Chapter 27 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Evidence/Provenance)	Consulted	Informed
Approve test acceptance authority/independence	Accountable Authority Class (to be assigned)	Evidence Assurance Owner (to be assigned)	Assurance/Audit (NCIE-008, independent per Ch.6 SoD)	NCIE-016 downstream owner

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-27-1: Confirm assurance independence, test acceptance authority and evidentiary standards for production readiness.

Traceability: NCIE-003 acceptance criteria; NCIE-008 assurance; NCIE-009 test-status discipline; NCIE-016 downstream.

Human Review Register & Open Evidence/Audit Decisions

Consolidate every unresolved Evidence/provenance/audit decision exposed across Chapters 1-27 and Chapter 29 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008/009 governance and security layers, or the approved NCIE-010 Skeletal v0.1 does not itself settle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008/009 Human Review registers, plus Evidence/provenance/audit-specific unresolved items from Chapters 1-27.

28.0 Salient Things to Note

Deduplicate identical upstream governance/security decisions; create NCIE-010 items only for distinct Evidence/provenance/audit questions. Blocking scope must be precise.

28.1 Purpose

This register exists so that no unresolved Evidence/provenance/audit decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-27 below was extracted programmatically from that chapter's own Human Review Focus section (§28.4). Chapter 28's own item (HR10-28-1) and Chapter 29's own item (HR10-29-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

28.2 Summary

Table 35. Human Review Register summary (mechanically computed).

Classification	Count
INSTITUTIONAL CONFIRMATION REQUIRED	25
LEGAL/POLICY CONFIRMATION REQUIRED	1
RESOLVED BY UPSTREAM BASELINE	1
SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	2
Total	29
— Blocking	22
— Non-Blocking	7

28.3 Complete Register

Table 36. Complete Human Review Register.

Review ID	Source Chapter	Question	Classification	Blocking Scope	Upstream Cross-Reference	Status
HR10-1-1	Chapter 1	Confirm NCIE-010 acceptance authority and resolve any unresolved overlap with audit/assurance responsibilities	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm NCIE-010 acceptance authority and resolve any unresolved overlap with audit/assurance responsibilities	NCIE-008 HR8-10-1 (acceptance/chapter sign-off authority) — same institutional question, instantiated for NCIE-010; resolving HR8-10-1 for the	Open

		already defined in NCIE-008/009.		already defined in NCIE-008/009.	NCIE-series pattern resolves this item too.	
HR10-2-1	Chapter 2	Approve any institution-specific evidence admissibility principles not fixed upstream; does not block the doctrine already stated above in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Open
HR10-3-1	Chapter 3	Confirm any evidence-class extensions that require a canonical-data amendment (routed to NCIE-003) rather than NCIE-010-only treatment.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm any evidence-class extensions that require a canonical-data amendment (routed to NCIE-003) rather than NCIE-010-only treatment.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-4-1	Chapter 4	Approve Evidence qualification authorities and criteria not already established upstream.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve Evidence qualification authorities and criteria not already established upstream.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-5-1	Chapter 5	Confirm which acquisition classes require heightened human attestation or independent corroboration.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm which acquisition classes require heightened human attestation or independent corroboration.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-6-1	Chapter 6	Approve integrity assurance requirements by Evidence class without inventing specific algorithms prematurely.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve integrity assurance requirements by Evidence class without inventing specific algorithms prematurely.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-7-1	Chapter 7	Confirm which Evidence classes require formal custody attestation or dual-control.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm which Evidence classes require formal custody attestation or dual-control.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-8-1	Chapter 8	Approve correction authority and confirm whether any Evidence classes require human validation before a corrected version becomes current.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve correction authority and confirm whether any Evidence classes require human validation before a corrected version becomes current.	N/A — distinct Evidence/provenance/audit decision	Open
HR10	Chapter	Confirm	INSTITUTIONAL	Confirm	N/A — distinct	Open

-9-1	er 9	invalidation/withdrawal authority and the appeal/review process.	CONFIRMATION REQUIRED (Blocking)	invalidation/withdrawal authority and the appeal/review process.	Evidence/provenance/audit decision	
HR10-10-1	Chapter 10	Approve relationship vocabulary extensions and any rules for declaring independent corroboration; does not block the six relationship types already fixed in Table 13.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Open
HR10-11-1	Chapter 11	Confirm the governance for declaring a contradiction resolved versus merely superseded.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the governance for declaring a contradiction resolved versus merely superseded.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-12-1	Chapter 12	Confirm which annotation classes may be considered authoritative expert opinion and under what review; does not block ordinary non-authoritative annotation in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Open
HR10-13-1	Chapter 13	Approve which transformations may yield new Evidence versus remain non-Evidence analytical artifacts.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve which transformations may yield new Evidence versus remain non-Evidence analytical artifacts.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-14-1	Chapter 14	Confirm whether any AI-assisted acquisition/transformation workflows can produce Evidence and what human controls they require.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm whether any AI-assisted acquisition/transformation workflows can produce Evidence and what human controls they require.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-15-1	Chapter 15	Confirm expert-witness/analyst/validator contribution classes if institutionally required; does not block the generic attribution structure already defined in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Open

HR10-16-1	Chapter 16	Confirm retention horizon and authority for formal historical reconstruction where not fixed upstream.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm retention horizon and authority for formal historical reconstruction where not fixed upstream.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-17-1	Chapter 17	Confirm the minimum evidentiary package/authority for validating Findings where unresolved.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the minimum evidentiary package/authority for validating Findings where unresolved.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-18-1	Chapter 18	Approve any institution-specific provenance relationship vocabulary beyond the canonical baseline; does not block the baseline graph model already defined.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Open
HR10-19-1	Chapter 19	Confirm audit-event classes requiring heightened assurance or independent review.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm audit-event classes requiring heightened assurance or independent review.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-20-1	Chapter 20	Confirm heightened-audit access classes and who may review sensitive audit records.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm heightened-audit access classes and who may review sensitive audit records.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-21-1	Chapter 21	Confirm audit retention/access requirements for sensitive governance deliberation; does not block ordinary governance-decision audit capture in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Open
HR10-22-1	Chapter 22	Confirm which external actions require additional evidentiary receipts or dual confirmation.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm which external actions require additional evidentiary receipts or dual confirmation.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-23-1	Chapter 23	Confirm special handling classes for highly sensitive	SECURITY/SOVEREIGNTY CONFIRMATION	Confirm special handling classes for highly sensitive	N/A — distinct Evidence/provenance/audit decision	Open

		Evidence/provenance/audit records.	REQUIRED (Blocking)	Evidence/provenance/audit records.	e/audit decision	
HR10-24-1	Chapter 24	Confirm retention authorities/periods and legal-hold ownership where unresolved.	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm retention authorities/periods and legal-hold ownership where unresolved.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-25-1	Chapter 25	Confirm discoverability rules for restricted/withdrawn Evidence and cross-investigation discovery.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm discoverability rules for restricted/withdrawn Evidence and cross-investigation discovery.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-26-1	Chapter 26	Confirm the authority for declaring audit/provenance reconstruction complete after material gaps.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the authority for declaring audit/provenance reconstruction complete after material gaps.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-27-1	Chapter 27	Confirm assurance independence, test acceptance authority and evidentiary standards for production readiness.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm assurance independence, test acceptance authority and evidentiary standards for production readiness.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-28-1	Chapter 28	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-010 v1.0 may be approved or may instead remain open under an interim design default.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-010 v1.0 may be approved or may instead remain open under an interim design default.	N/A — distinct Evidence/provenance/audit decision	Open
HR10-29-1	Chapter 29	Approve the NCIE-010 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-010 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as	RESOLVED BY UPSTREAM BASELINE (Non-Blocking)	Non-blocking — does not gate chapter-level Evidence/provenance/audit operation	N/A — distinct Evidence/provenance/audit decision	Resolved by upstream baseline

historical record, not
a live open item.

28.4 Mechanical Verification

27 of these 29 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-27 rather than assembled by hand. The remaining 2 items are HR10-28-1 (this chapter's own item) and HR10-29-1 (Chapter 29's own item), each defined once and rendered identically in both this table and their own chapter's Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

28.5 Upstream Deduplication (Directive Item 57)

One item, HR10-1-1, restates an institutional question NCIE-008 has already opened as HR8-10-1 (acceptance authorities for each NCIE-series document and chapter-level sign-off); Table 36 cross-references it rather than treating it as an independent open question, per directive item 57. No other NCIE-010 item was found to duplicate an existing NCIE-008/009 governance or security question — each remaining item concerns an Evidence/provenance/audit-specific decision (a qualification criterion, an integrity requirement, a retention authority, a test/acceptance authority) that NCIE-008/009 do not themselves resolve.

28.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific control default, threshold, ownership assignment or acceptance authority — rather than implying the whole of NCIE-010 or the whole approved upstream chain is blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's Evidence/provenance/audit operation in the interim.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR10-28-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-010 v1.0 may be approved or may instead remain open under an interim design default.

Traceability: All chapters above (mechanically reconciled register); NCIE-008/009 Human Review registers.

Traceability, Acceptance & Downstream Handover

Close traceability to the approved NCIE-001-009 foundation and the approved NCIE-010 Skeletal v0.1; define acceptance criteria and the downstream handover to NCIE-011/014/015/016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Approved NCIE-001-009 foundation; NCIE-010 Skeletal v0.1.

29.0 Salient Things to Note

Skeleton approval authorizes expansion only. Architecture approval does not prove implementation, evidentiary admissibility, operational audit completeness or test execution — §29.4's acceptance gate states this explicitly rather than leaving it implied.

29.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-010 Skeletal v0.1 itself. No chapter states an Evidence/provenance/audit requirement without an identifiable source.

Table 37. Full traceability matrix.

NCIE-010 Chapter	Primary Upstream Authority	Relationship
Chapter 1	Master Controlled Register v2.0; NCIE-002 Ch.4; NCIE-003 Ch.17/25; NCIE-006/007	Mandate/architecture boundary
Chapter 2	NCIE-001; NCIE-003; NCIE-005/007	Evidence & provenance doctrine
Chapter 3	NCIE-003 Ch.17/25	Evidence Object Model & canonical interfaces
Chapter 4	NCIE-003 Evidence lifecycle step 01; NCIE-001	Source registration & qualification
Chapter 5	NCIE-001; NCIE-003; NCIE-004/011	Acquisition & capture provenance
Chapter 6	NCIE-003 Ch.17; NCIE-009	Integrity metadata/hashing/tamper evidence
Chapter 7	NCIE-003 Ch.17 §17.6; NCIE-006/007	Chain of custody architecture
Chapter 8	NCIE-003; NCIE-006	Evidence versioning/correction/supersession
Chapter 9	NCIE-003 Evidence lifecycle step 06; NCIE-006/007	Invalidation/withdrawal/reliability
Chapter 10	NCIE-003 Ch.17 relationship model	Evidence relationships/support/contradiction
Chapter 11	NCIE-001/005; NCIE-006; NCIE-007	Contradiction preservation & evidence gaps
Chapter 12	NCIE-003 Evidence lifecycle step 04; NCIE-005; NCIE-007	Annotation/commentary/expert interpretation
Chapter 13	NCIE-002/003; NCIE-004; NCIE-009	Derived Evidence & transformation lineage
Chapter 14	NCIE-005/006/007; NCIE-003	AI/Agent/ARGUS provenance boundary
Chapter 15	NCIE-003/005; NCIE-008	Human contribution & attribution provenance

Chapter 16	NCIE-003; NCIE-006; NCIE-007	Temporal semantics & Decision-Time state
Chapter 17	NCIE-001; NCIE-003 Ch.16/17; NCIE-005	Evidence-to-Finding/Decision traceability
Chapter 18	NCIE-003 Ch.17/19/25; NCIE-002	Provenance graph & lineage architecture
Chapter 19	NCIE-003 Audit Event model; NCIE-006/007/009	Audit event architecture
Chapter 20	NCIE-001; NCIE-009; NCIE-008	Authorization & access audit
Chapter 21	NCIE-008; NCIE-009	Governance decision & approval audit
Chapter 22	NCIE-004 Tool Gateway; NCIE-007; NCIE-009	Tool/workflow/external-system provenance
Chapter 23	NCIE-003; NCIE-006; NCIE-009; NCIE-008	Security/privacy/classification of records
Chapter 24	NCIE-006; NCIE-008; NCIE-014 (downstream)	Retention/legal hold/archival/disposition
Chapter 25	NCIE-006; NCIE-009; NCIE-003	Evidence search/discovery/retrieval provenance
Chapter 26	NCIE-004/006/009; NCIE-007	Audit & provenance reconciliation/recovery
Chapter 27	NCIE-003; NCIE-008; NCIE-009; NCIE-016 (downstream)	Verification/assurance/defensibility testing
Chapter 28	All chapters above; NCIE-008/009 Human Review registers	Consolidated Human Review Register
Chapter 29	Approved NCIE-001-009; NCIE-010 Skeletal v0.1	Traceability/acceptance/handover

29.2 Traceability, Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 29 sequential 'NCIE-010 — Chapter N Expansion' labels, N = 1..29, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 69). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 29 chapters (directive item 70), rerun independently against this final document rather than assumed from the build script — the exact discipline the NCIE-009 Chapter-27 correction established.

29.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-010 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 68); none is omitted, and none is conceptually inverted.

29.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-010 v1.0 requires: (a) 29/29 chapter integrity confirmed; (b) 29/29 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 28 either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority; (e) the §27.1 verification/assurance test suite specified with no test falsely marked executed or passed; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any control default, threshold, product/algorithm choice or ownership assignment proposed above. Architecture acceptance under this chapter does not resolve open

NCIE-008/009 governance or security decisions, does not prove implementation, does not certify evidentiary admissibility, and does not establish that audit/defensibility tests have been executed.

29.5 Downstream Handover Register

NCIE-010 does not preempt detailed downstream specification. The table below states what is handed over and what remains downstream.

Table 38. Downstream handover register.

Downstream Document	What NCIE-010 Hands Over
NCIE-011 (detailed interfaces)	Evidence Object, custody-event and audit-event interface contracts (Ch.3, 7, 19) for detailed API design; no interface design is specified here.
NCIE-014 (physical storage)	Logical retention/legal-hold/archival/disposition requirements (Ch.24); no physical storage system or database is selected here.
NCIE-015 (operations)	Evidence/provenance/audit operational responsibilities (RACI/control-ownership tables throughout) for handoff into operational runbooks; no runbook is specified here.
NCIE-016 (detailed testing)	The 8 verification/assurance test classes specified in §27.1 (Table 33); detailed test cases, execution and automation remain downstream in NCIE-016 — this document specifies the tests, not their execution.

29.6 Readiness and Final Disposition

NCIE-010 Evidence, Provenance & Audit Architecture v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed control default, threshold, ownership assignment and product/algorithm choice deferred in Chapters 1-27 remains a proposal pending the Chapter 28 Human Review Register; every §27.1 verification/assurance test remains SPECIFIED, not executed, pending NCIE-016.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR10-29-1: Approve the NCIE-010 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-010 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-009 approved foundation; approved NCIE-010 Skeletal v0.1.